

Received 17 April 2024, accepted 15 May 2024, date of publication 23 May 2024, date of current version 3 June 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3404878

RESEARCH ARTICLE

Flexible and Fully Quantized Lightweight TinyissimoYOLO for Ultra-Low-Power Edge Systems

JULIAN MOOSMANN¹, (Graduate Student Member, IEEE),
HANNA MÜLLER¹, (Graduate Student Member, IEEE),
NICKY ZIMMERMAN², (Graduate Student Member, IEEE),
GEORG RUTISHAUSER¹, (Graduate Student Member, IEEE),
LUCA BENINI^{1,3}, (Fellow, IEEE),
AND MICHELE MAGNO¹, (Senior Member, IEEE)

¹Department of Information Technology and Electrical Engineering, ETH Zürich, 8092 Zürich, Switzerland

²IDSIA, Università della Svizzera Italiana, 6900 Lugano, Switzerland

³Department of Electrical, Electronic and Information Engineering, University of Bologna, 40126 Bologna, Italy

Corresponding author: Julian Moosmann (julian.moosmann@pbl.ee.ethz.ch)

This work was supported by armasuisse Science and Technology.

ABSTRACT This paper deploys and explores variants of TinyissimoYOLO, a highly flexible and fully quantized ultra-lightweight object detection network designed for edge systems with a power envelope of a few milliwatts. With experimental measurements, we present a comprehensive characterization of the network's detection performance, exploring the impact of various parameters, including input resolution, number of object classes, and hidden layer adjustments. We deploy variants of TinyissimoYOLO on state-of-the-art ultra-low-power extreme edge platforms, presenting a detailed comparison on latency, energy efficiency, and their ability to efficiently parallelize the workload. In particular, the paper presents a comparison between a RISC-V-based parallel processor (GAP9 from GreenWaves Technologies) with and without use of its on-chip hardware accelerator, an ARM Cortex-M7 core (STM32H7 from ST Microelectronics), two ARM Cortex-M4 cores (STM32L4 from ST Microelectronics and Apollo4b from Ambiq), and a multi-core platform aimed at edge AI applications with a CNN hardware accelerator (MAX78000 from Analog Devices). Experimental results show that the GAP9's hardware accelerator achieves the lowest inference latency and energy at 2.12 ms and 150 μ J respectively, which is around 2x faster and 20% more energy efficient than the next best platform, the MAX78000. The hardware accelerator of GAP9 can even run an increased resolution version of TinyissimoYOLO with 112×112 pixels and 10 detection classes within 3.2 ms, consuming 245 μ J. We also deployed and profiled a multi-core implementation on GAP9 at different core voltages and frequencies, achieving 11.3 ms with the lowest-latency and 490 μ J with the most energy-efficient configuration. With this paper, we demonstrate the flexibility of TinyissimoYOLO and prove its detection accuracy on a widely used detection dataset. Furthermore, we demonstrate its suitability for real-time ultra-low-power edge inference.

INDEX TERMS YOLO, ML, computer vision, object detection, hardware accelerator, microcontroller, quantization, quantization-aware training, network deployment, network deployment evaluation.

The associate editor coordinating the review of this manuscript and approving it for publication was Md. Arafatur Rahman¹.

I. INTRODUCTION

With the widespread adoption of edge devices, particularly microcontroller unit (MCU) based nodes, the Internet of Things (IoT) is revolutionizing various domains including